

19. Graph exposure: Sand

Completed 100k study | six faithful models, three paired seeds | all 1,080 full-H314 rollouts complete

Base-only and mixed-graph models train from initialization on all 1,000 training trajectories. Paired arms share initialization, sampled frames and noise. Mix expands half of training examples with a random quarter of optional annulus pairs; the native capped base graph remains intact.

Inference policy	Base-only mean +/- SD	Mixed graph mean +/- SD	Mix minus base mean +/- SD [seed signs]
Base	0.06785 +/- 0.06378	0.06781 +/- 0.06218	-0.00004 +/- 0.00263 [-/-/+]
Dense	0.12934 +/- 0.14479	0.12602 +/- 0.14751	-0.00332 +/- 0.00296 [-/-/-]
Random25	0.08298 +/- 0.08643	0.07240 +/- 0.06802	-0.01058 +/- 0.01861 [-/-/+]
Speed25	0.08085 +/- 0.07898	0.09071 +/- 0.09794	+0.00986 +/- 0.01898 [-/+/+]
Cached risk25	0.09714 +/- 0.10755	0.10631 +/- 0.11951	+0.00917 +/- 0.01201 [+/+/+]
Relative-velocity RMS25	0.08782 +/- 0.09293	0.08507 +/- 0.08888	-0.00275 +/- 0.00498 [-/-/+]

Full-horizon position MSE; each seed equally averages all 30 trajectory means. Means and sample SDs describe three seeds. Signs are ordered seeds 0/1/2; a minus favors mixed training. All six policies and all required outcomes are retained.

Useful training effect, conditional expansion. Exposure lowers observed-test error under all five expanded policies in every seed; dense full-rollout error also improves in every seed. Yet native base beats both random25 and dense in every seed of both arms. Learning to use an expanded graph does not imply that expansion outperforms the native graph.

Residual ranking does not transfer through feedback

On matched observed test histories, the risk-minus-random deficit narrows from +4.76 +/- 3.80 to +1.87 +/- 1.04 (position-coordinate MSE x 10⁻⁹), but risk remains worse in every seed of both arms. In autonomous rollouts, exposure worsens cached-risk error in every seed and widens its risk-minus-random gap by +0.01975 +/- 0.03062, also adverse in every seed. Observed risk uses the preceding observed base graph; autonomous risk caches its own selected-graph scores.

Completion still leaves physical and cost limitations

Mixed random25 places 10.47% of particles outside the metadata box by more than 1e-6 (truth: 0.0618%), with mean trajectory-maximum excursion 0.3244 (truth: 0.0007954). Across all policies, mean outside fractions fall after exposure, but maximum excursions increase in every paired seed. These are geometric diagnostics, not conservation tests.

Mean committed-call time increases for all six policies; random25 changes from 14.06 +/- 0.90 to 14.70 +/- 1.66 seconds/case. Calls include native parity. Fixed policy order, shared hosts and evolving geometries prevent an isolated speedup claim.

Coverage and scope

All 3,648 mixed-stage cells complete: 1,080 autonomous outcomes, 900 observed validation histories, 900 observed test histories and 768 clean-validation frames. None failed or remained uncompleted. Sand is an exploratory, distinct native-CUDA lineage; earlier strict CPU/CUDA gradient checks failed, so CPU equivalence and a controlled material effect are not claimed.

Source: completed Sand100k paired summary and audited saved-array/scalar evidence; manuscript appendix, "Sand: graph exposure and placement." The preceding 20 report pages preserve their dated snapshots. This page contains no Goop3D numerical claim.